COGNIFYZ TECHNOLOGIES

DATA SCIENCE INTERNSHIP

LEVEL 2 REPORT

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Domain: Data Science

Tools Used: Python, Pandas, Matplotlib, Jupyter Notebook, Anaconda

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Introduction

The objective of Level 2 is to perform advanced analysis on the restaurant dataset by examining table booking services, online delivery services, price range patterns, and feature engineering techniques. These analyses provide deeper insights into customer preferences and restaurant characteristics.

Task 1: Table Booking and Online Delivery Analysis

Objective

To determine the percentage of restaurants offering table booking and online delivery services and analyze their relationship with restaurant ratings and price ranges.

Findings

- Percentage of restaurants offering Table Booking: 12.12%
- Percentage of restaurants offering Online Delivery: 25.66%
- Average Rating of restaurants with Table Booking: 3.44
- Average Rating of restaurants without Table Booking: 2.56

Observations

- Online delivery services are more common than table booking services.
- Only a small percentage of restaurants offer table booking facilities.
- Restaurants with table booking tend to receive higher ratings.
- Customer convenience features appear to positively influence restaurant ratings.

Screenshots

(Insert Table Booking Percentage Output)

(Insert Online Delivery Percentage Output)

(Insert Average Rating Comparison Output)

(Insert Table Booking Bar Chart)

(Insert Online Delivery Bar Chart)

Task 2: Price Range Analysis

Objective

To analyze restaurant price ranges and determine their relationship with customer ratings.

Findings

Restaurant Distribution by Price Range

- Price Range 1: 4444 restaurants
- Price Range 2: 3113 restaurants
- Price Range 3: 1408 restaurants
- Price Range 4: 586 restaurants

Average Rating by Price Range

- Price Range 1: 2.00
- Price Range 2: 2.94
- Price Range 3: 3.68
- Price Range 4: 3.82

Observations

- Price Range 1 is the most common category.
- Higher-priced restaurants generally receive better ratings.
- Average customer satisfaction increases with price range.

Conclusion

There is a positive relationship between restaurant price range and customer ratings. Premium restaurants tend to achieve higher average ratings compared to lower-priced restaurants.

Screenshots

(Insert Price Range Distribution Output)

(Insert Average Rating by Price Range Output)

(Insert Price Range Analysis Chart)

Task 3: Feature Engineering

Objective

To create additional features from existing columns and convert categorical variables into numerical representations.

Features Created

Name Length

Calculated the length of each restaurant name.

Address Length

Calculated the length of each restaurant address.

Has Table Booking

Encoded categorical values:

- Yes = 1
- No = 0

Has Online Delivery

Encoded categorical values:

- Yes = 1
- No = 0

Observations

- Feature engineering improves dataset usability for machine learning models.
- Encoded variables allow algorithms to process categorical information effectively.
- Additional engineered features can improve predictive model performance.

Conclusion

New features were successfully extracted from the dataset, and categorical variables were converted into numerical formats. The transformed dataset is now suitable for advanced analytics and predictive modeling tasks.

Screenshots

(Insert Feature Engineering Output Screenshot)

(Insert Encoded Features Screenshot)

Overall Conclusion

Level 2 tasks were successfully completed using Python and data analysis libraries. The analysis revealed important insights regarding customer convenience services, restaurant pricing strategies, and feature engineering techniques. The dataset was transformed into a more informative and machine-learning-ready format.

The objectives of all Level 2 tasks were successfully achieved.